MIL-C-3363C(GL)
16 September 1982
SUPERSEDING
MIL-C-3363B
11 September 1970

MILITARY SPECIFICATION

CHEST, MEDICAL EQUIPMENT MAINTENANCE AND REPAIR TOOL KIT

This specification is approved for use by the Natick Research and Development Laboratories, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of a wooden tool chest with a removable tray.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-H-111	-	Hardware, Builders'; Shelf and Miscellaneous
FF-S-111	-	Screw, Wood
QQ-S-781	-	Strapping, Steel, and Seals
RR-C-271	-	Chains and Attachments, Welded and Weldless and Roller Chain
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5140

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TT-E-529	-	Enamel, Alkyd, Semi-Gloss
TT-P-636	-	Primer Coating, Alkyd, Wood and Ferrous Metal
MMM-A-181	-	Adhesives Phenol, Resorcinol or Melamine Base
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	-	Strapping, Nonmetallic (and Connectors)
PPP-T-60	-	Tape, Waterproof, Packaging
PPP-T-76	-	Tape, Packaging, Paper, (for Carton Sealing)

MILITARY

MIL-S-13518	-	Wood Preservative: Tetrachlorophenol and Pentachlorophenol, Surface Sealing Compound

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-1188	-	Commercial Packing of Supplies and Equipment
MS 35647	-	Padlocks, Key Operated

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

10-1-529	-	Chest, Medical Equipment Maintenance and Repair Tool Kit; Assembly

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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OTHER GOVERNMENT DOCUMENTS

UNITED STATES DEPARTMENT OF COMMERCE

Product Standard PS-1 - Softwood Plywood - Construction and Industrial

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 109 - Steel, Carbon, Cold Rolled Strip

D 2016 - Standard Methods of Test for Moisture Content of Wood

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed by technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

* 3.2 Materials. Materials shall conform to the requirements specified herein and in the subsidiary documents applicable thereto (see 6.5).

* 3.2.1 Steel strip. The strip steel used in the chest shall confrom to ASTM A 109.

* 3.2.2 Plywood. Plywood used in the construction of the bottom of the tray shall conform to group 1, exterior type, grade B-B medium density overlay of Product Standard PS 1.

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* 3.2.3 Lumber. Lumber for solid wooden parts of the chest shall be any of the following species: white ash, beech, birch, hickory, hard maple, oak or pecan. Lumber shall be uniformly dried to an average moisture content of from 8 to 12 percent at the time of machining (see 4.4.1.1). The minimum board width shall be 2 inches. Slope of grain of machined components shall not exceed one inch in fifteen. Machined components shall be free of decay, splits, wane, warp and unsound knots.

3.2.4 Adhesives.

3.2.4.1 High temperature setting resin. High temperature setting resin adhesive shall conform to the requirements of type III of MMM-A-181, except that the shear strength by compression loading requirement and the curing cycle requirement shall not apply.

3.2.4.2 Room temperature and intermediate temperature setting resin. Room temperature and intermediate temperature setting resin adhesive shall conform to any grade of types I or II of MMM-A-181.

3.2.5 Screws. Screws shall conform to FF-S-111, of the types and sizes shown on Drawing 10-1-529.

3.2.6 Hinge hasps. Hinge hasps shall conform to type F1405E of FF-H-111. Holes in the hasps shall be countersunk to accept flat countersunk head rivets.

3.2.7 Padlocks. Padlocks shall conform to MS 35647-3.

3.2.8 Chain. Chain shall conform to Trade No. 40, type II, class 3, steel sash chain of RR-C-271. The required chain has approximately 22 links per foot with a nominal metal link thickness of 0.042.

3.2.9 Primer. Primer shall conform to TT-P-636.

3.2.10 Enamel. Enamel shall conform to TT-E-529. Unless otherwise specified (see 6.2), the color for the chest shall be olive drab conforming to Color No. 24087 of FED-STD-595. The color for identification marking shall be black conforming to Color No. 37038 of FED-STD-595.

3.2.11 Wood sealer. Wood sealer shall conform to MIL-S-13518.

3.3 Design. The chest shall consist of a body, a lid, partitions, a removable tray, and shall be equipped with metal hardware and handles, all as shown on Drawing 10-1-529.

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3.4 Construction. Construction shall conform to the requirements shown on Drawing 10-1-529 and as specified herein. When panels are made from two or more boards, they shall be joined by means of tongue-and-grooving and gluing. All exposed wooden surfaces shall be evenly and smoothly dressed. The chest shall be assembled to true 90 degree angles. The lid shall fit down on the body with not more than 1/32 inch opening at any point. Steel binding shall fit flush to wood without ripples. Rivets shall be fabricated from rivet-grade steel wire, and shall have flat countersunk heads. Handles and hinge hasps shall be attached to the chest with rivets specified above, and applied so as not to interfere with the proper positioning of the removable tray. Corner braces shall be fabricated from low carbon sheet or strip steel specified in 3.2.1 having a thickness of 0.058 inch plus or minus 0.007. Hinge bodies shall be fabricated from low carbon sheet or strip steel specified in 3.2.1 having a thickness of 0.075 inch plus or minus 0.010 inch. Hinge pins shall be made of 3/16 inch diameter steel wire, shall be neatly headed, and shall fasten the leaves of the hinges securely. All wood screws shall be screwed into place and not driven. Predrilled holes for screws shall be no larger than the minor diameter of the screw and shall not exceed 3/8 inch in wood penetration.

3.5 Finish.

3.5.1 Wood surfaces preparation. After assembly of the wooden components of the chest, but before hardware components are attached, the inside and outside wooden surfaces, including partitions and tray, shall be sanded smooth and given one coat of wood sealer specified in 3.2.12.

3.5.2 Metal surface preparation. The surface of steel parts, excluding screws, rivets, padlock and chain, shall be given a cleaning and surface preparation treatment conforming to method 1, type 1 of TT-C-490.

3.5.3 Priming.

3.5.3.1 Lid and body. Prior to attachment of hardware, a coat of the primer specified in 3.2.10 shall be applied to all inside and outside surfaces of the lid and body. Forced drying is permitted, provided a surface temperature of 140 degrees Fahrenheit (F) is not exceeded and the relative humidity of the surrounding air is maintained between 30 to 50 percent. The coating shall level out to a dry uniform film thickness without objectionable orange peel, runs, wrinkles, drops, streaks or areas of no film, and shall be free from grit or coarse particles. All parts shall be thoroughly dry prior to assembly of hardware.

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3.5.3.2 Metal parts. All metal parts, excluding screws, rivets, padlock and chain, after the treatment specified in 3.5.2 and before assembly, shall be given a coat of primer specified in 3.2.10.

3.5.4 Enameling. Prior to the attachment of the hardware a coat of enamel specified in 3.2.11 shall be applied to all inside and outside surfaces of the lid and body. All parts shall be thoroughly dry prior to assembly of hardware. After complete assembly of the chest, all accessible surfaces (including the hardware components), shall be given a second coat of enamel. The enamel shall level out to produce a smooth and uniform film without objectionable orange peel, runs, wrinkles, drops, streaks or area of no film and shall be free of grit or coarse particles. All surfaces of the lid and body shall be thoroughly dry before stacking or packing.

* 3.6 Marking for identification. In accordance with MIL-STD-130, each chest shall be marked with the enamel specified in 3.2.11, in the location shown on Drawing 10-1-529.

3.7 Workmanship. The chest shall have no splintered, jagged or sharp wood or metal edges. Nails shall be driven so as not to split the wooden members. Rivets shall not be over driven or protrude so as to cause damage to the chest.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2.), it shall be examined for the defects classified in table III and the dimensions specified. Any visual defect or dimension not within specified requirements shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

* 4.4.1.1 Moisture content test of lumber. Lumber shall be tested prior to machining for moisture content in accordance with ASTM D 2016 to determine conformance to 3.2.3. The lot shall be all lumber offered for inspection at one time. The sample unit shall be one piece of lumber. Three determinations shall be taken on each sample unit and the results shall be the average of the three determinations. The inspection level shall be S-3 with an AQL of 2.5 expressed in terms of defects per hundred units.

* 4.4.1.2 Intermediate inspection. Prior to assembly, machined plywood and wood components shall be examined for defects listed herein. The sample unit shall be one machined component. The lot shall be all the machined components offered for inspection at one time. The inspection level shall be I and the acceptable quality level (AQL) shall be 4 defects per hundred units.

TABLE I. Intermediate examination

Examine	Defect
Wood components	Not species specified. Slope of grain exceeds 1 inch in 15. Evidence of decay, splits, wane, warp, and unsound knots.

* 4.4.2 In-process inspection. Examination shall be made during the manufacturing process to determine that the components are being fabricated in accordance with specified requirements. Whenever nonconformance is noted, the affected items and processes shall be corrected.

TABLE II. In-process inspection

Examine	Requirement paragraph
Fabrication and dressing of wooden chest surfaces	3.4
Fitting of lid to body	3.4
Application of steel binding to chest	3.4
Attachment of hardware components to chest	3.4

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TABLE II. In-process inspection (cont'd)

Examine	Requirement paragraph
Predrilling of holes and method for securing screws	3.4
Preparation of wood surfaces	3.5.1
Preparation of steel component surfaces	3.5.2
Priming of lid and body	3.5.3.1
Forced drying of primer coat (if applicable)	3.5.3.1
Priming of hardware components prior to assembly	3.5.3.2
Application of enamel coatings	3.5.4

* 4.4.3 End item examination. The end item shall be examined in accordance with 4.4.3.1 and 4.4.3.2. The lot size shall be expressed in units of one chest. The sample unit shall be one chest.

* 4.4.3.1 Visual examination. The chest shall be examined for defects classified in table III. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for total (major and minor combined) defects.

TABLE III. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Color of finish not as specified.		X
	Finish is not smooth or uniform e.g., has runs, streaks, orange peel, wrinkles, drops.		X
	Area of no film.		X
	Wet or tacky finish.	X	
	Not free of grit or coarse particles.	X	

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TABLE III. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Workmanship	Chest not assembled to true 90 degree angles.	X	
	Lid does not fit down into body.	X	
	Steel binding not flush to wood without ripples.	X	
	Rivet heads not countersunk	X	
	Handles and hinges not attached as specified.	X	
	Component missing.	X	
	Component not securely attached or joined as specified.	X	
	Lid does not align evenly with body when in the closed position - openings exceed 1/32 inch at any point.	X	
	Splintered, jagged or sharp wooden or metal edge.	X	
Fastening devices	Rivet not properly set, peened or clinched.		X
	Rivet overdrawn or protruding from chest.		X
	Screw not tight.		X
	One rivet or screw missing.		X
	More than one rivet or screw missing.	X	
	Nail protrudes from or splits wooden member.	X	
Identification marking	Missing, illegible, incorrect, wrong location, wrong color, or not as specified.		X

* 4.4.3.2 Dimensional examination. Inspection shall be made of the completed chests for compliance with dimensions specified. Any dimension that is not within the specified requirements shall be classified as a defect. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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* 4.4.4 Packaging inspection. An examination shall be made to determine that preservation, packing and marking comply with section 5 requirements. Defects shall be scored as specified in table IV. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination of closure defects listed in table IV shall be made on shipping containers fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-3 and the AQL expressed in terms of defects per hundred units, shall be 6.5.

TABLE IV. Packaging defects

Examine	Defect
Markings	Omitted; incorrect, illegible; of improper size, location, sequence, or method of application.
Materials	Component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as loose strapping, strap protector inadequate or missing.

* 4.4.5 Palletization examination. An examination shall be made to determine that palletization is in compliance with section 5 requirements. Defects shall be scored as specified in table V. The sample unit shall be one palletized unit load, fully packaged. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

TABLE V. Palletization defects

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limit.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. The tray shall be immobilized in place within each chest in a manner as to preclude movement or damage while in transit. Each chest shall be closed and the lid secured with the padlock provided. The keys shall be secured to the lock with not less than 0.072 inch diameter, commercially available annealed wire, and then further secured with pressure sensitive tape conforming to PPP-T-76 or PPP-T-60.

* 5.2 Packing. Packing shall be level A or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Each chest, preserved as specified in 5.1.1, shall be protected with a solid fiberboard cap tray completely covering the top and extending down not less than 8 inches on all sides and ends of the chest. The cap tray shall be fabricated from fiberboard conforming to type SF, class domestic, grade 200 of PPP-F-320. The cap tray shall be secured to the chest with flat steel strapping, 3/8 inch wide by 0.015 inch thick, conforming to type I, finish B of QQ-S-781 or type II or III nonmetallic strapping, having a minimum breaking strength of 390 pounds of PPP-S-760. Three straps shall be used to secure the cap tray to the chest: (a) one lengthwise strap shall be centered and secured to the chest and (b) two girthwise straps shall be equally speced and secured to the chest. The bottom edges of the chest shall be protected from indentation by the steel strapping at each point of contact with edge protectors. The edge protectors may be eliminated when nonmetallic strapping is used.

* 5.2.2 Commercial packing. Chests, preserved as specified in 5.1.1 shall be packed in accordance with MIL-STD-1188.

* 5.3 Palletization. When specified (see 6.2) chests packed as specified in 5.2 shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the chest is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.4 Marking. In addition to any special marking required by the contract, each chest and palletized unit load shall be marked in accordance with MIL-STD-129 or MIL-STD-1188 as applicable. All markings shall be printed or stenciled on the fiberboard cap tray. Markings shall not be applied directly to the surface of the chest. Service color marking and symbol are not required on containers when addressed to domestic establishments.

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* 5.4.1 Name of item. The following nomenclature shall be used for identification purposes: "Chest, Medical Equipment Manitenance and Repair Took Kit".

6. NOTES

6.1 Intended use. The chest is intended to be used to contain and protect selected tools and equipment required for maintenance and repair of medical service equipment.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this document.
(b) When a first article is required (see 3.1, 4.2 and 6.4).
(c) Color, if other than specified (see 3.2.11).
(d) Selection of the applicable levels of packing (see 5.2).
(e) When palletization is required (see 5.3).

6.3 This document covers the following item appearing in the Federal Supply catalogs:

FSN 5140-914-3845 - Chest, Medical Equipment Maintenance and Repair Tool Kit.

* 6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

* 6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.6 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-3363C Chest, Medical Equipment Maintenance And Repair Tool Kit

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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